WEEKLY ASSIGNMENT

Python Foundations

1. What is Python, and why is it popular among beginners and developers?

Python is a high-level, general-purpose programming language known for its simple and readable syntax. It is popular because it is relatively easy to learn, requires less code for many tasks, has a large collection of libraries, and is supported by a large community. It is used in web development, automation, data analysis, artificial intelligence, and machine learning.

2. What is the difference between a compiler and an interpreter?

A compiler translates a program into machine code or another lower-level form before execution, often producing a separate executable. An interpreter executes program instructions through a runtime, generally without first creating a standalone executable. In practice, many language implementations combine compilation and interpretation; Python commonly compiles source code to bytecode, which its virtual machine executes.

3. What are keywords in Python? List any five keywords.

Keywords are reserved words that have special meanings in Python's syntax and cannot be used as ordinary identifiers, such as variable names. Five examples are `if`, `else`, `for`, `while`, and `def`.

4. What is indentation, and why is it important in Python?

Indentation is the space at the beginning of a line of code. Python uses indentation to define code blocks, such as the body of an `if` statement, loop, or function. Consistent indentation makes code readable and incorrect indentation can cause an `IndentationError` or change the program's structure.

5. What is a comment? Explain single-line and multi-line comments.

A comment is text in source code intended for people to read; Python ignores it when executing the program. A single-line comment begins with #, for example `# This is a comment`. For notes spanning several lines, each line can begin with #. Triple-quoted strings (`'''...'''` or `"""..."""`) are sometimes used as multi-line notes, but technically they are string literals, not comments.

```
# This is a single-line comment
# This is the first line
# This is the second line
```

6. What is a variable? State the rules for naming variables in Python.

A variable is a name that refers to a value or object in a program. Python variable names must begin with a letter or underscore; may contain letters, digits, and underscores; cannot begin with a digit; cannot contain spaces or special characters such as `@` or `-`; are case-sensitive; and cannot be Python keywords. For example, `student_name` and `marks2` are valid names, while `2marks` is invalid.

```
student_name = "Asha"
marks2 = 85
```

7. What is dynamic typing in Python?

Dynamic typing means that a variable's type is determined at runtime from the value assigned to it. You do not need to declare the type in advance, and the same variable name can later refer to a value of a different type. For example:

```
x = 10 # x refers to an integer
x = "hello" # x now refers to a string
```

8. What is the purpose of the input function?

The `input()` function reads information entered by the user through the keyboard. It returns the entered value as a string. If a number is needed, the string can be converted using functions such as `int()` or `float()`.

```
name = input("Enter your name: ")
```

9. What is the purpose of the print function?

The `print()` function displays text, variable values, or the results of expressions in the output. It is useful for showing information to users and checking program results while debugging.

```
print("Hello, world!")
```

10. How can you check the data type of a value in Python?

Use the built-in `type()` function to find the type of a value. For example, `type(10)` returns `<class 'int'>`, while `type('hello')` returns `<class 'str'>`.

```
value = 10
print(type(value)) # <class 'int'>
```